Compound percentages, simultaneous equations and quadratic graphs

The national "practical and interdisciplinary" lessons, carrying three Cambridge Stage 9 topics the programme does not otherwise reach.

LESSON 74–76 | 3 × 40 MIN | ALGEBRA 8, PRACTICAL PROBLEMS

STAGE 9 · 3.3, 4.2, 10.2 [CAMBRIDGE INSERT]

LESSON CLOCK

5' | 12' | 8' | 13' | 2'

Warm-up	5 min
Explanation	12 min
Interactive	8 min
Practice	13 min
Homework	2 min

LEARNING OBJECTIVES

- Calculate repeated percentage change with a multiplier.
- Solve a pair of simultaneous linear equations by substitution and by elimination.
- Plot the graph of a quadratic and read its roots and vertex.
- Apply all three to practical situations.

TEXTBOOK

National	pp. 159–164
Cambridge	Learner's Book pp. 66–73, 91–95, 218–224
Workbook	Workbook 3.3, 4.2, 10.2

01 Explanation

Lesson 74 — compound percentage change

THE MULTIPLIER

An increase of $p\%$ multiplies by $1 + \frac{p}{100}$; a decrease multiplies by $1 - \frac{p}{100}$. For n repetitions, raise the multiplier to the power n:

$$final = initial \times (multiplier)^n$$

A salary of 2 000 000 so'm rising 8% a year for 3 years: $2\ 000\ 000 \times 1.08^3 \approx 2\ 519\ 424$.

NOT THE SAME AS ADDING THE PERCENTAGES

> Three years at 8% is **not** 24%. It is $1.08^3 \approx 1.2597$, that is 25.97% — the extra comes from the growth compounding on itself.

A 20% rise followed by a 20% fall is $1.2 \times 0.8 = 0.96$ — a 4% **loss**, not a return to the start. This is the single most useful fact in the topic.

Lesson 75 — simultaneous equations

Substitution. Make one unknown the subject of one equation and put it into the other.

> $y = 2x + 1$ and $3x + y = 11 \Longrightarrow 3x + 2x + 1 = 11 \Longrightarrow x = 2, y = 5$

Elimination. Add or subtract the equations so that one unknown disappears — multiplying one or both first if you need matching coefficients.

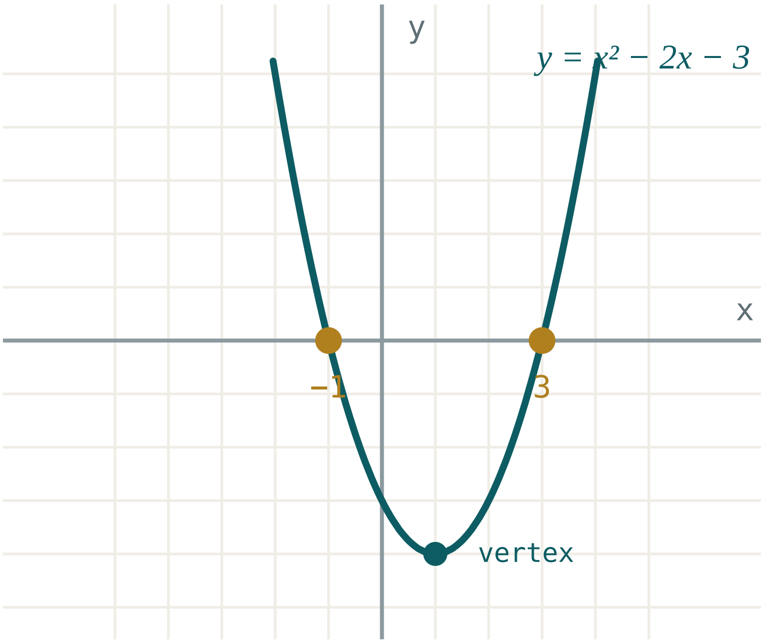
> $2x + 3y = 12$ and $2x - y = 4$: subtract $\Longrightarrow 4y = 8 \Longrightarrow y = 2, x = 3$

Graphically the solution is where the two lines cross. Parallel lines mean no solution; the same line twice means infinitely many.

Always **check in both equations** — a pair that satisfies only one is not a solution.

Lesson 76 — the graph of a quadratic

To draw $y = ax^2 + bx + c$, build a table of values, plot, and join with a smooth curve — never with straight segments.



The parabola crosses the x-axis at the roots and turns at the vertex.

WHAT TO READ OFF THE GRAPH

- **Roots** — where the curve meets the x-axis.
- **y-intercept** — the value c.
- **Axis of symmetry** — the vertical line $x = -\frac{b}{2a}$.
- **Vertex** — on that line; the lowest point if $a > 0$, the highest if $a < 0$.

$a > 0$ gives a curve opening upwards (a “valley”); $a < 0$ opens downwards (a “hill”). The graph is a picture of everything the discriminant told you.

02 Worked examples

EXAMPLE 1 **A population of 50 000 grows 3% a year. Find it after 4 years.**

1. $multiplier = 1.03$

   A 3% increase.

2. $50\ 000 \times 1.03^4$

   Four years.

3. $= 50\ 000 \times 1.1255$

4. $\approx 56\ 275$

ANSWER

$\approx 56\ 300$ people

EXAMPLE 2 **Solve $3x + 2y = 16$ and $x - y = 3$**

1. $x = y + 3$

   Make x the subject of the second.

2. $3(y + 3) + 2y = 16$

   Substitute.

3. $5y = 7, y = 1.4$

4. $x = 4.4$

   Check: $3(4.4) + 2(1.4) = 16$ ✓

ANSWER

$x = 4.4, y = 1.4$

EXAMPLE 3 ***For $y = x^2 - 2x - 3$, find the roots, the y-intercept and the vertex.***

1. $x^2 - 2x - 3 = 0 \Longrightarrow (x - 3)(x + 1) = 0$

   Roots 3 and -1.

2. $y\text{-intercept: } c = -3$

   Put $x = 0$.

3. $\text{axis: } x = -\dfrac{-2}{2} = 1$

4. $y = 1 - 2 - 3 = -4$

   Vertex $(1; -4)$.

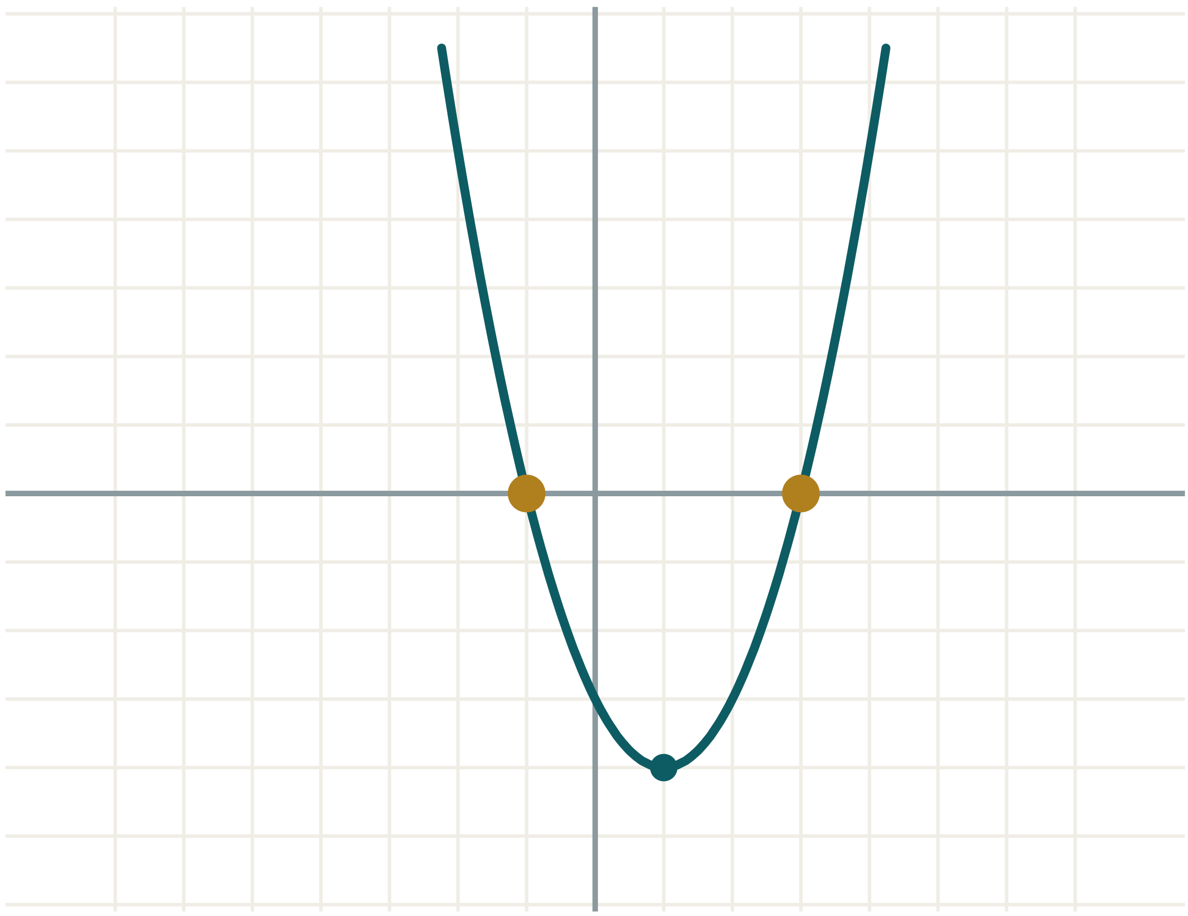
ANSWER

Roots $-1, 3$; intercept -3; vertex $(1; -4)$

03 Interactive model

Change a, b and c and let the class predict the vertex before the readout shows it.

Reading a parabola



a **1** | b **-2** | c **-3** | $D = b^2 - 4ac$ **16** | roots **two** | x_1 **-1** | x_2 **3** | $x_1 + x_2$ **2** | $x_1 \cdot x_2$ **-3**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Compound percentage	Murakkab foiz	Сложный процент
Multiplier	Ko‘paytiruvchi	Множитель
Increase / decrease	Ortish / kamayish	Увеличение / уменьшение
Simultaneous equations	Tenglamalar sistemasi	Система уравнений
Substitution method	O‘rniga qo‘yish usuli	Метод подстановки
Elimination method	Qo‘shish usuli	Метод сложения
Table of values	Qiymatlar jadvali	Таблица значений
Vertex of a parabola	Parabola uchi	Вершина параболы
Axis of symmetry	Simmetriya o‘qi	Ось симметрии

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

A 10% rise then a 10% fall leaves you with:

- the same
- 1% more
- 1% less
- 10% less

Three years at 5% multiplies by:

- 1.15
- 1.05^3
- 3×1.05
- 0.95^3

The axis of symmetry of $y = x^2 - 6x + 5$ is:

- $x = 3$
- $x = -3$
- $x = 6$
- $x = 5$

Two simultaneous equations whose lines are parallel have:

- one solution
- no solution
- two solutions
- infinitely many

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. *Increase 200 by 10%*

   ANSWER *220*

2. *Decrease 200 by 10%*

   ANSWER *180*

3. *Write the multiplier for a 7% increase*

   ANSWER *1.07*

4. *Write the multiplier for a 15% decrease*

   ANSWER *0.85*

5. Solve $x + y = 7$ and $x - y = 1$

   ANSWER $x = 4, y = 3$

6. Solve $y = 2x$ and $x + y = 9$

   ANSWER $x = 3, y = 6$

7. Find the y-intercept of $y = x^2 + 3x - 4$

   ANSWER -4

Medium · the standard question

7 PROBLEMS

1. *A population of 50 000 grows 3% a year. Find it after 4 years.*

   ANSWER $\approx 56\ 300$

2. *A car worth 30 000 000 so‘m loses 12% a year. Find its value after 3 years.*

   ANSWER $\approx 20\ 440\ 000$

3. *A 20% rise is followed by a 20% fall. Find the overall change.*

   ANSWER *4%* decrease

4. Solve $3x + 2y = 16$ and $x - y = 3$

   ANSWER $x = 4.4, y = 1.4$

5. Solve $2x + 3y = 12$ and $2x - y = 4$

   ANSWER $x = 3, y = 2$

6. Find the roots and vertex of $y = x^2 - 2x - 3$

   ANSWER Roots $-1, 3$; vertex $(1; -4)$

7. Find the axis of symmetry of $y = 2x^2 - 8x + 1$

   ANSWER $x = 2$

Hard · stretch and reason

7 PROBLEMS

1. *A sum grows to 1.331 times its value in 3 years at a constant rate. Find the rate.*

   ANSWER $r^3 = 1.331$, so $r = 1.1$ — 10% a year.

2. *A price falls 25% then rises 25%. Find the overall change.*

   ANSWER $0.75 \times 1.25 = 0.9375$ — a 6.25% loss.

3. *Solve $4x - 3y = 11$ and $3x + 2y = 4$*

   ANSWER $x = 2, y = -1$

4. *Two numbers add to 30 and differ by 8. Find them, using simultaneous equations.*

   ANSWER 19 and 11

5. *Sketch $y = -x^2 + 4x$ and give its roots and vertex.*

   ANSWER Roots 0 and 4; vertex $(2; 4)$; opens downwards.

6. *For which values of c does $y = x^2 - 4x + c$ touch the x-axis?*

   ANSWER $D = 0$ gives $c = 4$.

7. *A shop raises a price by 10% each year for 2 years, then cuts it by 21%. Show the price returns to the start.*

   ANSWER $1.1^2 \times 0.79 = 1.21 \times 0.79 \approx 0.956$ — close, but a 4.4% loss; an exact return needs a cut of 17.36%.

07 Homework

Homework — 6 problems

Algebra 8, pp. 159–164, and Cambridge Learner’s Book 3.3, 4.2, 10.2.

1. *A town of 80 000 grows 2% a year. Find its population after 5 years.*
2. *A machine worth 12 000 000 so‘m loses 15% a year. Find its value after 4 years.*
3. *Solve $5x + 2y = 21$ and $3x - y = 4$*
4. *Solve $2x + y = 10$ and $x - 3y = -9$*

5. *Draw the graph of* $y = x^2 - 4x + 3$ *for* x *from* -1 *to* 5*, and read off the roots.*
6. *Find the vertex and the axis of symmetry of* $y = x^2 + 6x + 5$